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ULSAN, Korea, Republic of

WMO: **471520**

Lat: **35.593N** Lon: **129.352E** Elev: **45** StdP: **14.67** Time Zone: **9.00 (E09)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	21.4	24.3	-8.2	3.6	28.8	-4.3	4.4	31.1	18.0	29.3	16.2	32.4	6.9	340	0.371	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	8	11.5	91.8	77.0	89.5	76.5	87.1	75.5	79.5	86.8	78.5	85.4	77.4	84.3	7.4	140	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	77.8	144.5	82.9	76.5	138.6	82.4	75.4	133.1	81.9	43.2	86.9	42.1	85.3	41.0	84.4	83.5	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	15.5	13.4	11.8	DB	16.3	95.2	3.9	2.6	13.5	97.1	11.2	98.6	9.0	100.1	6.2	101.9	(4)
(5)				WB	12.5	80.5	3.4	1.7	10.0	81.7	8.1	82.7	6.2	83.6	3.7	84.9	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		58.5	36.7	40.1	47.3	56.8	65.2	71.0	78.1	79.6	71.4	62.5	51.3	40.4	(6)
(7)		DBStd		15.54	5.95	6.14	6.08	5.87	5.02	4.38	5.40	4.71	4.40	4.92	6.32	6.37	(7)
(8)		HDD50		1195	412	282	126	12	0	0	0	0	0	1	60	302	(8)
(9)		HDD65		3722	876	698	548	254	60	4	0	0	2	106	411	763	(9)
(10)		CDD50		4287	1	4	43	215	472	631	871	916	642	388	100	4	(10)
(11)		CDD65		1337	0	0	0	7	67	185	406	451	194	27	0	0	(11)
(12)		CDH74		10445	0	0	0	49	416	984	3657	4325	939	74	1	0	(12)
(13)		CDH80		3347	0	0	0	3	66	198	1323	1570	185	2	0	0	(13)
(14)	Wind	WSAvg		4.8	5.2	5.2	5.5	5.2	4.7	4.3	4.7	4.7	4.6	4.4	4.5	5.1	(14)
(15)	Precipitation	PrecAvg		50.7	1.3	1.6	2.8	4.3	4.1	6.9	8.3	8.7	6.9	2.5	2.1	0.9	(15)
(16)		PrecMax		81.1	4.7	5.0	7.8	8.8	7.9	21.7	24.4	27.5	26.0	11.6	6.5	4.8	(16)
(17)		PrecMin		27.3	0.0	0.0	0.1	0.9	0.4	0.4	1.3	0.9	1.1	0.0	0.0	0.0	(17)
(18)		PrecStd		13.7	1.1	1.2	1.7	2.1	2.0	4.4	4.9	5.8	5.1	2.4	1.8	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	62.3	69.5	79.3	85.6	87.9	94.2	95.5	88.7	79.0	71.6	61.0		(19)
(20)			MCWB	47.0	54.5	53.7	60.7	63.6	70.2	77.8	77.6	74.6	65.9	58.5	51.6		(20)
(21)		2%	DB	51.9	57.6	65.4	74.6	81.6	84.9	91.8	92.4	84.7	76.0	67.7	56.8		(21)
(22)			MCWB	42.4	47.4	51.4	57.5	62.1	69.6	76.9	77.3	72.4	64.3	57.2	47.2		(22)
(23)		5%	DB	48.7	53.8	61.9	71.1	78.5	82.1	89.6	90.1	81.8	73.7	64.7	53.9		(23)
(24)			MCWB	40.2	44.2	50.1	55.5	61.5	68.9	76.6	77.1	71.6	63.3	55.5	45.1		(24)
(25)		10%	DB	46.2	50.4	58.6	68.0	75.7	79.6	87.1	87.8	79.2	71.4	62.0	51.0		(25)
(26)			MCWB	38.5	41.8	48.5	55.1	60.9	68.3	76.2	76.5	70.3	61.7	53.4	43.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.9	57.5	59.3	65.8	70.8	75.7	81.0	81.0	78.0	72.3	63.5	54.4		(27)
(28)			MCDB	54.0	60.9	63.5	69.0	75.9	81.3	88.9	89.0	83.6	74.9	66.8	57.2		(28)
(29)		2%	WB	45.0	50.4	55.7	62.4	67.4	74.2	79.5	79.8	75.8	68.8	60.3	49.7		(29)
(30)			MCDB	48.9	55.2	61.3	70.8	74.0	79.6	86.8	87.3	81.5	72.3	65.2	55.0		(30)
(31)		5%	WB	42.2	45.7	52.5	59.6	65.2	72.5	78.3	78.9	74.2	66.0	57.4	46.8		(31)
(32)			MCDB	46.5	51.9	59.2	66.5	72.7	78.2	85.5	85.8	79.4	70.9	62.7	52.0		(32)
(33)		10%	WB	39.8	43.1	49.3	57.4	63.6	70.5	77.2	78.0	72.0	63.5	54.9	44.1		(33)
(34)			MCDB	45.0	48.9	57.0	63.7	71.7	76.7	84.8	84.7	77.0	69.4	60.4	49.9		(34)
(35)	Mean Daily Temperature Range		MDBR	14.8	15.6	16.2	17.2	16.3	12.9	11.1	11.5	12.2	15.4	16.4	15.7		(35)
(36)		5% DB	MCDBR	18.0	20.5	21.4	23.1	21.8	17.9	15.2	15.4	15.9	17.9	19.4	19.8		(36)
(37)			MCWBR	12.0	13.2	12.3	11.2	9.1	7.0	4.7	4.8	6.5	8.7	11.0	12.9		(37)
(38)		5% WB	MCDBR	14.6	17.6	17.8	17.1	16.4	12.7	13.2	12.6	12.3	13.5	15.4	16.8		(38)
(39)			MCWBR	11.7	13.5	13.0	11.1	9.3	6.6	5.2	5.0	6.3	8.5	11.0	12.9		(39)
(40)	Clear-Sky Solar Irradiance	taub			0.412	0.468	0.548	0.581	0.591	0.630	0.600	0.555	0.497	0.468	0.434	0.404	(40)
(41)		taud			2.062	1.952	1.788	1.765	1.791	1.771	1.903	2.016	2.120	2.082	2.085	2.096	(41)
(42)		Ebn at Noon			237	238	229	231	231	221	227	235	242	236	228	231	(42)
(43)		Edh at Noon			41	51	65	70	69	71	62	54	46	44	39	37	(43)
(44)	All-Sky Solar Radiation	RadAvg			849	1037	1350	1575	1724	1533	1422	1443	1195	1113	860	776	(44)
(45)		RadStd			70	127	135	125	176	142	221	171	131	109	110	51	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	N/A	(47)

Nomenclature: See separate page